

Indefinite Integrals of Trigonometric Functions

1. $\frac{d}{dx}(\sin ax) = a \cos ax \longrightarrow \int \cos ax \, dx = \frac{1}{a} \sin ax + C$
2. $\frac{d}{dx}(\cos ax) = -a \sin ax \longrightarrow \int \sin ax \, dx = -\frac{1}{a} \cos ax + C$
3. $\frac{d}{dx}(\tan ax) = a \sec^2 ax \longrightarrow \int \sec^2 ax \, dx = \frac{1}{a} \tan ax + C$
4. $\frac{d}{dx}(\cot ax) = -a \csc^2 ax \longrightarrow \int \csc^2 ax \, dx = -\frac{1}{a} \cot ax + C$

Substitution Rule for Indefinite Integrals Let $u = g(x)$, where g' is continuous on $[a, b]$, and let f be continuous on the range of g . Then

$$\int f(g(x))g'(x) \, dx = \int f(u) \, du.$$

General Slicing Method. Suppose a solid object extends from $x = a$ to $x = b$ and the cross section of the solid perpendicular to the x -axis has an area given by a function A that is integrable on $[a, b]$. The volume of the solid is

$$V = \int_a^b A(x) \, dx$$

Disk Method About the x -Axis. Let f be continuous with $f(x) \geq 0$ on the interval $[a, b]$. If the region R bounded by the graph of f , the x -axis, and the lines $x = a$ and $x = b$ is revolved about the x -axis, the volume of the resulting solid of revolution is

$$V = \int_a^b \pi f(x)^2 \, dx.$$

Arc length. Let f have a continuous first derivative on the interval $[a, b]$. The length of the curve from $(a, f(a))$ to $(b, f(b))$ is

$$L = \int_a^b \sqrt{1 + f'(x)^2} \, dx.$$

Helpful Formulas

Sections 7.1, 7.2, 7.3, and 8.1

Derivative of the Inverse Function. Let f be differentiable and have an inverse on an interval I . If x_0 is a point of I at which $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and

$$\left(f^{-1}\right)'(y_0) = \frac{1}{f'(x_0)}, \text{ where } y_0 = f(x_0).$$

Helpful Formulas: Sections 8.1, 8.2, 8.3

Formulas for Derivatives

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$ (Chain rule)
3. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
4. $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$
5. $\frac{d}{dx}(\sin ax) = a \cos ax$
6. $\frac{d}{dx}(\cos ax) = -a \sin ax$
7. $\frac{d}{dx}e^x = e^x$
8. $\frac{d}{dx}e^{u(x)} = u'(x)e^{u(x)}$
9. $\frac{d}{dx}(b^x) = b^x \ln b$
10. $\frac{d}{dx}(b^{u(x)}) = b^{u(x)}(\ln b)u'(x)$
11. $\frac{d}{dx} \ln x = \frac{1}{x}$
12. $\frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$, for $x > 0$

Formulas for Integration

1. $\int x^n dx = \frac{1}{n+1}x^{n+1} + C$; $n \neq -1$
2. $\int \cos ax dx = \frac{1}{a} \sin ax + C$
3. $\int \sin ax dx = -\frac{1}{a} \cos ax + C$
4. $\int e^{ax} dx = \frac{1}{a}e^{ax} + C$
5. $\int \frac{1}{x} dx = \ln|x| + C$
6. $\int b^x dx = \frac{1}{\ln b}b^x + C$ for $b > 0$ and $b \neq 1$
7. $\int \ln x dx = x \ln x - x + C$
8. $\int \tan x dx = \ln|\cos x| + C$
9. $\int \sec x dx = \ln|\sec x + \tan x| + C$
10. $\int f(g(x))g'(x) dx = \int f(u) du$ (Substitution rule)
11. $\int u dv = uv - \int v du$ (Integration by parts)

General Reduction Formulas

1. $\int \sin^n ax dx = -\frac{\sin^{n-1} ax \cdot \cos ax}{an} + \frac{n-1}{n} \int \sin^{n-2} ax dx$
2. $\int \cos^n ax dx = \frac{\cos^{n-1} ax \cdot \sin ax}{an} + \frac{n-1}{n} \int \cos^{n-2} ax dx$

Trigonometric Substitution Identities

1. $a^2 - x^2 \rightarrow a^2 \cos^2 \theta$ where $x = a \sin \theta$
2. $a^2 + x^2 \rightarrow a^2 \sec^2 \theta$ where $x = a \tan \theta$
3. $x^2 - a^2 \rightarrow a^2 \tan^2 \theta$ where $x = a \sec \theta$

Pythagorean Identity

$$4. \sin^2 \theta + \cos^2 \theta = 1$$

Half-Angle Formulas

$$5. \cos^2 \theta = \frac{1+\cos 2\theta}{2}; \quad \sin^2 \theta = \frac{1-\cos 2\theta}{2}$$

Double-Angle Identities

$$6. \sin 2\theta = 2 \sin \theta \cos \theta; \quad \cos 2\theta = 2 \cos^2 \theta - 1$$



images/Trig_Sub_reference_Triangle_a^2-x^2.png

1. $\int \tan^n ax \, dx = \frac{\tan^{n-1} ax}{a(n-1)} - \int \tan^{n-2} ax \, dx$

2. $\int \sec^n ax \, dx = -\frac{\sec^{n-2} ax \cdot \tan ax}{a(n-1)} + \frac{n-2}{n-1} \int \sec^{n-2} ax \, dx$